GDU Outlook

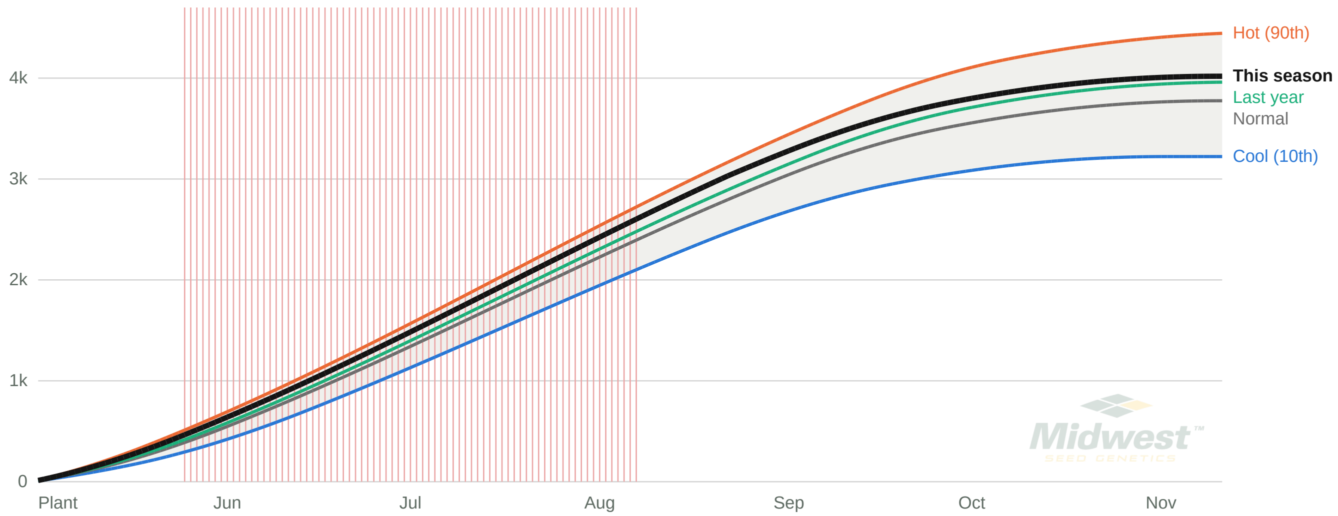
GDU accumulation

Missouri Valley, IA 51555 · planted May 1, 2026

2,602	**+208**	**99**
GDU ACCUMULATED	GDU AHEAD OF NORMAL	DAYS SINCE PLANTING

Observed through Aug 7, 2026, then the 16-day forecast, then projected.

GDU Accumulation



Vertical rules: red = the 75 days the high hit 86 °F, where heat above the cap added nothing.

Solid = observed through Aug 7 plus forecast through Aug 22. Dashed = projected.

Frost Risk

Oct 29	**Nov 7**	**Oct 30**
28 °F BY THIS DATE 1 YR IN 10	MEDIAN 28 °F FREEZE	MEDIAN 32 °F FROST

Read these as the LATE end of the range: a 6-to-15 mile grid cell averages away the radiative cooling that makes a low spot frost first, so against nearby station records these dates run 1 to 3 weeks late. GDU accumulation itself is within about 1% of station records.

How These Numbers Were Made

- GDU = (min(daily high, 86 °F) + max(daily low, 50 °F)) ÷ 2 - 50 — the modified base-50/86 method US seed companies rate hybrids on. A day below 50 °F counts 0, never a negative; heat above 86 °F adds nothing. Accumulation starts on the planting date itself.
- Normal, hot and cool are the 50th, 90th and 10th percentiles of accumulation across 30 complete years (1996–2025) at this exact grid point — an envelope, not a replay of any one year. Growth stages between Planting, Silks and Maturity are scaled from Iowa State's published ladder for a 2,700-GDU hybrid and are estimates.
- Temperatures: ERA5 reanalysis via Open-Meteo for history and the current season through Aug 7, 2026, plus Open-Meteo's 16-day forecast through Aug 22. Grid point 41.5644, -95.8913.
- No hybrid was entered, so no stage dates appear here — these curves are the heat itself. Add a hybrid in the app for silk and black layer predictions.
- GDU is a heat model, not a crop model. It knows nothing about drought, saturated soils, replant, hail, disease or nitrogen — any of which can move real silk and black layer dates well off these numbers.